

Disproof of conjecture 00000001067

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Verdict: FALSE

We disprove conjecture 00000001067 as stated. The refutation is unconditional, completely elementary, and finitely verifiable: for the prime $p = 19$, which satisfies $p \equiv 3 \pmod{4}$, the largest strong Sidon set in $\mathbb{F}_{19}$ has size 4, whereas $\lceil \sqrt{19} \rceil = 5$. The filed clause that the $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$ is an *exact* claim, and it is false. It fails already at $p = 11$ and also at $p = 43, 47, 59$.

1 The conjecture, quoted verbatim

From `conjectures/00000001067.md`:

English. Definition: A B_h set (unique h -fold sums). Conjecture: The maximal B_2 set in $\mathbb{F}_p$ has size $\lceil \sqrt{p} \rceil + O(1)$, and the $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$ (an exact B_2 result).

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim in Unicode in `README.md`, and the two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese is kept in the markdown copy rather than duplicated here.

The phrase “an exact B_2 result” in the last clause fixes the reading: the $O(1)$ term is asserted to *equal* 0, so the conjecture claims the exact identity

$$\max\{|S| : S \subseteq \mathbb{F}_p \text{ is a } B_2 \text{ set}\} = \lceil \sqrt{p} \rceil \quad \text{for every prime } p \equiv 3 \pmod{4}. \quad (1)$$

We refute (1).

2 Convention: B_2 means strong Sidon

Definition 1 (Strong Sidon / B_2 set). A subset $S \subseteq \mathbb{F}_p$ is a (*strong*) *Sidon set*, equivalently a B_2 set with unique 2-fold sums, if the sums $a + b$ with $a, b \in S$ and $a \leq b$ (repetition allowed) are pairwise distinct. Equivalently, the map $\{a, b\} \mapsto a + b$ from unordered pairs with repetition to $\mathbb{F}_p$ is injective.

This is the standard meaning of “unique 2-fold sums” and of B_2 : a B_h set has all h -fold sums (unordered, repetitions allowed) distinct, so a B_2 set is exactly a strong Sidon set as in Definition 1. In particular $2a = b + c$ with $b \neq c$ is forbidden, which is what distinguishes this from the *weak* Sidon reading discussed in Section 7.

3 The difference-count proof

Lemma 2 (Difference bound). *Let $S \subseteq \mathbb{F}_p$ be a strong Sidon set of size k . Then the $k(k-1)$ ordered differences $a-b$ with $a, b \in S$, $a \neq b$, are pairwise distinct and nonzero. Consequently*

$$k(k-1) \leq p-1.$$

Proof. Let $a \neq b$ and $c \neq d$, and suppose $a-b = c-d$. Then $a+d = c+b$. Both sides are sums of two elements of S with repetition allowed. If the multisets $\{a, d\}$ and $\{c, b\}$ are equal, then $(a, d) = (c, b)$: indeed $\{a, d\} = \{c, b\}$ means either $a = c$, $d = b$, or $a = b$, $d = c$, and the second alternative is excluded by $a \neq b$. In the first case $a-b = a-b$, i.e. $(a, b) = (c, d)$. If the two multisets differ, then $a+d = c+b$ is a collision between two distinct allowed sums, contradicting Definition 1. Hence the differences are pairwise distinct. They are nonzero because $a \neq b$. Finally, $\mathbb{F}_p$ has exactly $p-1$ nonzero elements, so $k(k-1) \leq p-1$. $\square$

Corollary 3. *There is no strong Sidon set of size 5 in $\mathbb{F}_{19}$, so the maximum is at most 4. More generally, at $p = 19$ one has $5 \cdot 4 = 20 > 18 = p-1$.*

Proof. If $|S| = 5$, Lemma 2 would give $5 \cdot 4 = 20 \leq 19-1 = 18$, a contradiction. $\square$

Proposition 4. *The maximum size of a strong Sidon set in $\mathbb{F}_{19}$ is exactly 4.*

Proof. The set $\{0, 1, 3, 7\}$ is strong Sidon: its ten unordered sums are

$$0, 1, 3, 7, 2, 4, 8, 6, 10, 14$$

(reading $\{0+0, 0+1, 0+3, 0+7, 1+1, 1+3, 1+7, 3+3, 3+7, 7+7\}$), i.e. the sorted list

$$0, 1, 2, 3, 4, 6, 7, 8, 10, 14,$$

which is pairwise distinct. Hence the maximum is at least 4; Corollary 3 gives at most 4. Therefore it equals 4. $\square$

Theorem 5. *Conjecture 00000001067 is false.*

Proof. At the prime $p = 19$ we have $p \equiv 3 \pmod{4}$ (indeed $19 = 4 \cdot 4 + 3$) and $\lceil \sqrt{19} \rceil = 5$, because $4^2 = 16 < 19 \leq 25 = 5^2$. By Proposition 4 the maximal strong Sidon set in $\mathbb{F}_{19}$ has size 4, not 5. Thus the exact identity (1) fails at $p = 19$, which refutes the “ $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$ ” clause. $\square$

Remark 6. $p = 11$ is an even smaller counterexample: $11 \equiv 3 \pmod{4}$, $\lceil \sqrt{11} \rceil = 4$, and Lemma 2 forbids $k = 4$ because $4 \cdot 3 = 12 > 10 = 11-1$; a maximum 3-set is $\{0, 1, 3\}$. We use $p = 19$ as the headline witness because the difference bound there is $5 \cdot 4 = 20 > 18 = p-1$.

4 Exhaustive verification at $p = 19$

The counting argument above is decisive, but the claim can also be checked directly. Enumerating all $\binom{19}{5} = 11628$ five-element subsets of $\mathbb{F}_{19}$ and testing the 15 unordered sums of each, one finds that *every* five-subset contains a repeated sum: there is no strong Sidon 5-set, confirming Corollary 3 without the difference bound. `reproduce.py` performs exactly this enumeration. It also reports that all 2129 strong Sidon subsets of $\mathbb{F}_{19}$ have size at most 4, and that the maximum is attained (a maximum 4-set is $\{0, 1, 3, 7\}$).

5 The full table

We computed the maximum strong Sidon size in $\mathbb{F}_p$ by exhaustive backtracking for $p = 7, 11, 19, 23, 31, 43, 47, 59$. Column “strong” is the true maximum; “ $\lceil \sqrt{p} \rceil$ ” is the conjectured exact value.

p	$p \bmod 4$	$\lceil \sqrt{p} \rceil$	max strong Sidon	holds?
7	3	3	3	yes
11	3	4	3	NO
19	3	5	4	NO
23	3	5	5	yes
31	3	6	6	yes
43	3	7	6	NO
47	3	7	6	NO
59	3	8	7	NO

The exact formula (1) happens to hold at $p = 7, 23, 31$ but fails at $p = 11, 19, 43, 47, 59$. This is what makes the conjecture plausible but false rather than obviously wrong: the values agree on the first few primes $p \equiv 3 \pmod{4}$ that one is likely to test, and only break later. For $p = 43, 47, 59$ the difference bound alone does not forbid the conjectured size: at $p = 43$ one has $7 \cdot 6 = 42 = 43 - 1$, at $p = 47$, $7 \cdot 6 = 42 \leq 46$, and at $p = 59$, $8 \cdot 7 = 56 \leq 58$; it is the exhaustive search (and, for $p = 43$, the Bruck–Ryser–Chowla theorem below) that rules the larger size out.

6 Cross-check at $p = 43$: Bruck–Ryser–Chowla

At $p = 43$ a strong Sidon set of size $k = 7$ would satisfy $k(k - 1) = 42 = p - 1$, so by Lemma 2 its 42 differences would be *all* the nonzero elements of $\mathbb{F}_{43}$; such a set is a perfect $(43, 7, 1)$ difference set. A (v, k, λ) difference set with $k - \lambda = 6$ gives a symmetric 2 -(v, k, λ) design, and a symmetric 2 -($n^2 + n + 1, n + 1, 1$) design is a projective plane of order n . Here $v = 43 = 6^2 + 6 + 1$ and $k = 7 = 6 + 1$, so a size-7 strong Sidon set in $\mathbb{F}_{43}$ would produce a projective plane of order 6.

The Bruck–Ryser–Chowla theorem states that if a projective plane of order n exists and $n \equiv 1, 2 \pmod{4}$, then n is a sum of two squares. Here $6 \equiv 2 \pmod{4}$, and 6 is not a sum of two squares (0, 1, 2, 4, 5 are representable below 6, but 6 is not). Hence no projective plane of order 6 exists, so no size-7 strong Sidon set in $\mathbb{F}_{43}$ exists. This is consistent with the computed maximum 6, and it independently certifies the failure of (1) at $p = 43$.

7 Alternative readings

We tried to save the conjecture by changing the reading; none succeeds.

Weak Sidon

Definition 7 (Weak Sidon). $S \subseteq \mathbb{F}_p$ is *weak Sidon* if the sums $a + b$ with $a, b \in S$, $a \neq b$ (distinct elements only) are pairwise distinct.

This is genuinely weaker: repetitions $2a$ are unconstrained. The weak maxima for the same primes are

p	$\lceil \sqrt{p} \rceil$	max weak Sidon	holds?
7	3	4	NO
11	4	5	NO
19	5	6	NO
23	5	6	NO
31	6	7	NO
43	7	8	NO
47	7	8	NO
59	8	9	NO

Under the weak reading the exact formula fails even more dramatically: at $p = 19$ the maximum is $6 > 5 = \lceil \sqrt{19} \rceil$ (for instance $\{0, 1, 2, 4, 7, 12\}$ works), and at $p = 11$ it is $5 > 4 = \lceil \sqrt{11} \rceil$ (for instance $\{0, 1, 2, 4, 7\}$). Note also that Lemma 2 is a *strong*-Sidon statement: the weak maximum set $\{0, 1, 2, 4, 7, 12\}$ in $\mathbb{F}_{19}$ has $6 \cdot 5 = 30$ ordered differences but $\mathbb{F}_{19}$ has only 18 nonzero elements, so its differences are certainly not distinct. The difference-count proof must not be applied to weak Sidon sets; for weak Sidon the correct counting bound is $\binom{k}{2} \leq p$, since the $\binom{k}{2}$ sums of distinct pairs must lie in $\mathbb{F}_p$.

“Maximal” as inclusion-maximal

If “maximal” is read as *inclusion-maximal* (a strong Sidon set that is not properly contained in another strong Sidon set) rather than maximum cardinality, nothing changes: every inclusion-maximal strong Sidon set in $\mathbb{F}_{19}$ has size 4. There are 1140 such sets, and all have size 4; none has size 5. So this reading does not rescue the $O(1) = 0$ clause either.

Asymptotic weakening

Only a substantially weakened asymptotic version survives. The classical upper bound for strong Sidon sets in $\mathbb{F}_p$ is $|S| \leq \sqrt{p} + O(1)$ (immediately from $k(k-1) \leq p-1$), so the asymptotic part of the conjecture is true and standard; what fails is precisely the exact “ $O(1)$ term = 0” clause, i.e. the assertion that the maximum equals $\lceil \sqrt{p} \rceil$ for $p \equiv 3 \pmod{4}$. Replacing the exact claim by $|S| = \lceil \sqrt{p} \rceil + O(1)$ would make it true but vacuous as a statement about the $O(1)$ term.

8 Reproducibility

The exhaustive computations are carried out by `reproduce.py` using the Python 3 standard library only:

```

$ python3 reproduce.py
... (checks, the strong/weak tables,
    difference bound,
    Bruck--Ryser cross-check)
PASS: all checks verified;
    conjecture 00000001067 is FALSE.

```

It computes the strong and weak maxima for all eight primes by exhaustive backtracking, verifies the difference bound on every maximum set, verifies the witness $\{0, 1, 3, 7\}$ by listing its sums, enumerates all $\binom{19}{5} = 11628$ five-subsets, checks the equivalence “strong Sidon $\Leftrightarrow$ differences distinct” on all subsets of $\mathbb{F}_7, \mathbb{F}_{11}$ and all subsets of size ≤ 6 of $\mathbb{F}_{19}, \mathbb{F}_{23}$, and reports the Bruck–Ryser–Chowla and inclusion-maximal facts. It prints **PASS** and exits 0 exactly when every check holds.

9 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only, `import Std`, no `Mathlib`, no `sorry`). Over the field $\mathbb{F}_{19}$ presented as `Fin 19`, the main statements are

```

theorem no_sidon5 :
  ∀ a b c d e : Fin 19,
    a < b → b < c → c < d → d < e
    → sidon5 a b c d e = false
theorem sidon4_witness :
  sidon4 0 1 3 7 = true
theorem count_obstruction :
  5 * 4 > 19 - 1
theorem ceil_sqrt19 :
  4 ^ 2 < 19 ∧ 19 ≤ 5 ^ 2
theorem conjecture_00000001067_false :
  (∃ a b c d : Fin 19,
    sidon4 a b c d = true)
  ∧ no_sidon5 ∧ ceil_sqrt19

```

Here `sidon5 a b c d e` decides whether the 15 unordered sums $a+a, \dots, e+e$ are pairwise distinct (strong Sidon), and `sidon4` is the analogous predicate for four elements. The proof of `no_sidon5` uses translation invariance of the strong Sidon property: replacing each element x by $x - a$ shifts every sum by $-2a$, an injection on $\mathbb{F}_{19}$, so it suffices to refute the case $a = 0$, which `decide` checks exhaustively; the general case is then reduced to it. The audit `Check.lean` reports no `sorryAx` and no `Lean.ofReduceBool` (the latter would signal `native_decide`, which is deliberately not used); the only axioms are the core `propext` and `Quot.sound`. See `lean4/README.md` for the full statement table and the honest axiom footprint.

Status against the submission rules

Rule 3 requires a LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic main.tex`.
- **PDF** — at `build/main.pdf`, produced by `tectonic --outdir build main.tex`.
- **Lean 4 project** — `lean4/` with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`,

```
lakefile.toml (library Main, no dependencies), Main.lean, Check.lean (#print axioms),  
and lean4/README.md. lake build exits 0 and the audit reports no sorryAx.
```